PROJECT RECOMMENDATIONS

1. Model Selection

This report provides recommendations based on the results obtained from the Logistic Regression and Gradient Boosting models developed to predict retail investor dormancy. The aim is to identify investors whose previous behaviour indicates a higher likelihood of becoming inactive, allowing the client to take appropriate retention action before dormancy occurs.

The modelling results indicated that Gradient Boosting achieved higher accuracy, F1-score, and ROC-AUC than Logistic Regression. Although Logistic Regression achieved a higher recall, it has recorded an accuracy of 61.66%, precision of 3.48%, recall of 85.90%, F1-score of 6.69% and ROC-AUC of 0.787 while Gradient Boosting achieved an accuracy of 77.12%, precision of 5.42%, recall of 80.77%, F1-score of 10.15% and ROC-AUC of 0.862.

Gradient Boosting accuracy is approximately ten percentage points higher than Logistic Regression, while its ROC-AUC increased from 0.787 to 0.862. The higher ROC-AUC indicates stronger discrimination between dormant and non-dormant investors across different classification thresholds. Its higher F1-score also indicates a better balance between identifying dormant investors and limiting incorrect positive predictions.

Overall, Gradient Boosting demonstrated stronger results and is therefore recommended because it provides a better balance between identifying dormant investors, avoiding false alarms and distinguishing dormant from non-dormant investors. This is consistent with the model description developed for the project, where Gradient Boosting was selected because it can capture nonlinear relationships and interactions between behavioural variables, while Logistic Regression serves as a simpler and more interpretable baseline (Ala'raj, Abbod and Majdalawieh, 2021).

2. Model Improvement

The first improvement should be hyperparameter tuning. The current Gradient Boosting model uses predefined settings for parameters such as learning rate, number of estimators, tree depth and regularisation. These parameters should be systematically tuned using cross-validation so that will help the model generalise better to new, unseen data (Ala'raj, Abbod and Majdalawieh, 2021). Grid Search or Randomised Search could be used to test alternative combinations of these parameters and identify a better-performing configuration.

Secondly, the classification threshold should be optimised. The current default threshold of 0.50 may not be appropriate for the client's business objective. Testing different thresholds such as 0.30, 0.40 and 0.60 could identify an appropriate threshold to balance the business cost of missing dormant investors against the cost of unnecessary retention actions. A lower threshold could increase the number of potential dormant investors identified, while a higher threshold could reduce false positives.

Third, additional class imbalance techniques should be investigated. Although the current Gradient Boosting model uses sample weights, techniques such as controlled oversampling, under sampling or alternative cost-sensitive learning could be tested. PR-AUC, recall and precision should remain important evaluation measures because the dormant class is relatively small.

In addition, the model's probability estimates should be assessed and calibrated. This is important if the client intends to use the model to assign investors risk scores or prioritise them into different intervention groups. Previous financial modelling research highlights the practical importance of calibrated probabilities when customer scores are used for decision-making (Ala'raj, Abbod and Majdalawieh, 2021).

Finally, model explainability should be incorporated. Feature importance and explainable AI techniques could identify which behavioural factors contribute most strongly to dormancy predictions. This would make the model easier for business stakeholders to understand and support more targeted interventions.

3. Adapting the model to SS1 data

The model will need to be adapted when the requested SS1 data from the digital investment platform becomes available. The project currently uses COFINFAD public dataset as a broader fintech dataset rather than a dedicated stock-investment platform dataset. The first adaptation should involve replacing the current transaction-based dormancy proxy with the client's actual operational definition of an inactive investor. The current research definition is based on whether a customer transacts during the future prediction period.

The second adaptation is the feature mapping process. The SS1 data should be transformed into the same variables used by the model, including transaction frequency, transaction value, trading frequency, active days and transaction-type behaviour. Investment specific variables, such as investment activity or trading patterns should be incorporated because they are more directly related to investor dormancy.

The dormancy target should also be redesigned. For example, in the current public data dormancy is defined as recently active customers who make no transaction during a subsequent 60-day period. For SS1, the business should determine whether dormancy should be defined using 30, 60, 90 or another period based on actual investor behaviour and business requirements.

The preprocessing should then be reapplied to the SS1 data, including treating missing values, encoding categorical values, numerical scaling where required, removal of identifiers and exclusion of variables that introduce data leakage. Performance should then be tested later on actual SS1 investor data.

4. Alignment with Related Literature

The findings align with the related literature reviewed in Part A. Muñoz-Guerrero, Ceballos and Trejos-Rojas (2026) demonstrate the value of combining transactional, demographic, product usage and digital engagement information for analysing fintech customer behaviour. The current study similarly demonstrates that transaction based behavioural variables can be used to develop predictive models for future inactivity.

The findings are also consistent with behavioural scoring research. Ala’raj, Abbod and Majdalawieh (2021) demonstrated that machine learning can be used to predict future customer financial behaviour and recommended the use of probability scores to support customer risk grouping. Their study evaluated several traditional and neural-network approaches using measures including accuracy, AUC, Brier score, calibration and McNemar's test.

The present results similarly show the value of evaluating probability quality in addition to accuracy. Gradient Boosting achieved lower Log Loss and Brier Score values, suggesting that its predicted probabilities provide useful additional information beyond a simple dormant/non-dormant classification.

However, the current findings also extend the literature by focusing on retail investor dormancy on digital investment platforms, rather than credit-card payment behaviour. The public dataset provides a useful proxy, but the use of SS1 platform data would provide a stronger test of whether these relationships apply specifically to retail investors.